

Guide for participating physicians

Thank you for taking part. This guide explains the study, how to install TimeStamp once, and what to do for each case. Please work as you normally would; the study is designed to add as little extra work as possible.

About the study

Most medical imaging AI is trained to predict the final diagnosis or report from an image. A physician does not reach a diagnosis in one step: you explore the image, focus on different regions, consider possible findings, gather evidence, update your interpretation and finally write the report. **This study models that diagnostic process itself**, not only the final outcome.



We will use a **world model** to learn how a physician's diagnostic state changes over time: from the image, what has been seen so far, mouse interaction and dictation, it learns how your interpretation evolves and what is likely to happen next. Instead of only *Image* → *Diagnosis*, it learns *Image* + *diagnostic history* → *current diagnostic state* → *next action or observation* → *updated state*. It aims to capture both:

- **Diagnostic reasoning:** how your interpretation changes as new evidence is observed.
- **Diagnostic behavior:** where you search and what you choose to inspect next.

The pilot: about 30 cases

This first pilot tests whether the recorded signals are enough to reconstruct meaningful diagnostic behavior. We want to find out whether we can:

- identify the stages of the diagnostic process;
- infer which finding or question you are investigating;
- predict which image region you will inspect next;
- predict which finding may be reported next;
- model how the evidence you gather leads to the final report;
- describe how diagnostic strategies differ between physicians.

Long-term goal

A world model of physician diagnostic behavior that captures how a diagnosis is reached, not only the diagnosis. It could help study search strategies, compare diagnostic behaviors, spot incomplete search patterns, predict next diagnostic actions, and build AI that supports physicians during diagnosis.

The pilot is a feasibility experiment for the data-collection protocol itself.

What is recorded for each case

Signal	How it is captured
Medical image	Which slide you have open in QuPath, and the region and zoom you are viewing. The image itself is not copied.
Mouse position and path	Automatic: your cursor position on the image about 10 times per second, plus zooms, pans and clicks.
Dictation	Your voice and an automatic transcript, with the time of every word.
Findings	Say each finding aloud as you identify it. Any annotations you draw in QuPath are also recorded.
Final report	Dictate your final report aloud at the end of the case.

Everything is synchronized on one clock, stays on this computer, and is only recorded while TimeStamp shows **Recording**. Nothing is recorded while paused. There is no screen or camera recording.

Install (once)

You need

- **QuPath 0.6.0**, opened once so its first-time setup is complete.
- **Windows 10 or 11 (x64), a Mac, or Linux**. Windows on ARM is not supported.
- **Internet for setup** (about 3.6 GB downloads once), **a microphone**, and **permission to install** (on a hospital computer, ask IT).
- The file `TimeStamp-Doctor-0.2.0-preview.20261005.zip`.

1 Note your QuPath folder, then quit QuPath

In QuPath, open **Preferences** and write down the **User directory**. Then quit QuPath; setup stops if QuPath is open.

2 Unzip the download

Windows: right-click the ZIP, choose **Extract All**. **Mac:** double-click the ZIP. Then open the

`TimeStamp-Doctor-0.2.0-preview.20261005` folder.

3 Run the installer for your computer

Windows

Double-click
`Install TimeStamp on Windows.bat`

Mac

Double-click
`Install TimeStamp.command`.
If blocked, Control-click it and
choose **Open**.

Linux

In a terminal in the folder:
`bash "Install TimeStamp on Linux.sh"`

If your computer blocks the installer, ask IT to approve it. Do not turn off security software.

4 Confirm the QuPath folder

Press **Enter** if the folder shown matches your User directory. If not, paste your folder and press **Enter**.

5 Wait, and say one sentence when asked

Keep the window open; the speech download takes the longest. When asked, say a sentence so setup can test the microphone; it then tells you whether the microphone works.

Setup is finished when you see: "TimeStamp is ready for the doctor."

If a download is interrupted, run the same installer again. Files already downloaded are reused.

Before your first case: a 1-minute test

Open QuPath and allow microphone access. Choose **Extensions** → **TimeStamp Extension** → **Open Clinical Session Recorder**, then ⚙ **Settings** → **Test microphone**. Record a few sentences on any slide (no patient information), move the mouse around, choose **Finish & review**, then **Save Session**. Send that saved test folder to the study coordinator so they can confirm everything was recorded.

For each case

- 1 Open the slide and the recorder**
Open the case slide in QuPath. Open the recorder: **Extensions** → **TimeStamp Extension** → **Open Clinical Session Recorder**.
- 2 Start recording, then read the slide as you normally would**
Choose **Start Recording** and wait until it says **Recording**. Navigate with the mouse as usual. **Think aloud**: say what you are looking at, what you notice and what you are considering.
- 3 Say each finding as you identify it**
For example: “Here, tumor invades the muscularis propria.” Annotations you draw in QuPath are recorded too, but are not required.
- 4 Dictate your final report**
When you have reached your conclusion, say your final report aloud, as you would dictate it.
- 5 Finish & review, then save**
Choose **Finish & review** and wait for the final transcript (it takes about as long as the recording). Correct any wrong words, especially numbers and negations. Then choose **Save Session**, name it with the **case ID only** (for example `case-07` ; never patient names or record numbers), tick **Include audio recording**, and save it in the study folder.

Need a break? **Pause** stops recording; **Resume** continues the same case. Choose **Finish & review** only when the case is complete.

If something goes wrong

What you see	What to do
“Please close QuPath completely”	Quit QuPath, then run the installer again.
“This package supports Windows x64 only”	This computer is Windows on ARM, which is not supported. Contact the study coordinator.
Windows mentions a missing DLL	Ask IT to install the Microsoft Visual C++ 2015–2022 x64 runtime, then run the installer again.
TimeStamp is not in the Extensions menu	Quit and reopen QuPath. Check the User directory, then run the installer again and enter that folder.
“No microphone signal detected”	Sound stopped arriving. Check the microphone or headset connection; the warning clears when sound returns.
The final transcript takes too long	Choose Cancel final pass · keep saved audio ; later use More → Retry final transcription .
QuPath closed before you saved	Reopen the recorder and use More → Recover an unsaved session .

Getting help

In the recorder, choose **More** → **Copy support information** and send it to the study coordinator with the step that failed and the exact message. Never send patient information in emails or screenshots.

TimeStamp is a research preview for supervised use and is not clinically validated. Recording and transcription run on this computer; audio is not sent to any service. Working recordings stay in the QuPath User directory under `timestamp/recordings` until removed by IT, even when not saved. Follow your organization’s policy for storing study data.